

Conjecture 00000000716: the sign in the p -adic reflection formula is not a function of $p \bmod 4$

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1 The conjecture and the readings refuted

Statement (English). *Conjecture: The correction term in $\Gamma_p(a) \cdot \Gamma_p(1-a) = \pm 1$ is an explicit function of $p \bmod 4$ (the sign of the p -adic reflection formula).* The Chinese text says the same.

Here Γ_p is Morita's p -adic gamma function, p an odd prime, $a \in \mathbb{Z}_p$. The text claims that the sign $\Gamma_p(a)\Gamma_p(1-a)$ is a function of $p \bmod 4$. We refute exactly two readings.

- (A) *The sign depends only on $p \bmod 4$:* there is $f : \mathbb{Z}/4 \rightarrow \mathbb{Z}$ with $\Gamma_p(a)\Gamma_p(1-a) = f(p \bmod 4)$ for every odd prime p and every $a \in \mathbb{Z}_p$. We prove more: for every odd prime p , the map $a \mapsto \Gamma_p(a)\Gamma_p(1-a)$ is not constant on $\mathbb{Z}_p$.
- (B) *For each fixed integer a , the sign is a function of $p \bmod 4$* (the function may depend on a). We refute it for $a = 4$: there is no $s : \mathbb{Z}/4 \rightarrow \{\pm 1\}$ with $\Gamma_p(4)\Gamma_p(-3) = s(p \bmod 4)$ for all odd primes p . Indeed the product is $\equiv -1 \pmod{3}$ for $p = 3$ and $\equiv 1 \pmod{7}$ for $p = 7$, and $3 \equiv 7 \pmod{4}$.

Reading not refuted. If the conjecture is read only at the point $a = 1/2$, it is *true*: by the reflection formula quoted below, $\Gamma_p(1/2)^2 = (-1)^{(p+1)/2}$, because the representative of $1/2$ in $\{1, \dots, p\}$ is $(p+1)/2$. This is a function of $p \bmod 4$. The statement writes a general a , so we do not adopt this reading, but a reviewer who does would find the conjecture true at $a = 1/2$. We make no claim about rational a other than integers.

Known formula (context only, not used in the proof). Wikipedia, “ p -adic gamma function” (https://en.wikipedia.org/wiki/P-adic_gamma_function), states: $\Gamma_p(x)\Gamma_p(1-x) = (-1)^{x_0}$, “where x_0 is the first digit in the p -adic expansion of x , unless $x \in p\mathbb{Z}_p$, in which case $x_0 = p$ rather than 0.” So the sign depends on $x \bmod p$, not on $p \bmod 4$. Our Lean proof does not use this formula; it computes the needed values directly.

2 Definitions

Definition 1 (Morita's product). For $n \in \mathbb{N}$ put $\text{mor}_p(n) = (-1)^n \prod_{0 \leq j < n, p \nmid j} j \in \mathbb{Z}$. The factor $j = 0$ never occurs, since $p \nmid 0$; so this is $(-1)^n \prod_{0 < j < n, p \nmid j} j$.

Definition 2 (Morita's p -adic gamma function). Wikipedia (same page) defines it as “the unique continuous function of a p -adic integer x (with values in $\mathbb{Z}_p$) such that $\Gamma_p(x) = (-1)^x \prod_{0 < i < x, p \nmid i} i$ for positive integers x ”.

The definition presupposes that such a function exists. We construct it. For $x \in \mathbb{Z}_p$ let $\text{appr}(x, k) \in [0, p^k)$ be the natural number with $x \equiv \text{appr}(x, k) \pmod{p^k}$ (Mathlib's `PadicInt.appr`), and set

$$\Gamma_p(x) = \lim_{k \rightarrow \infty} \text{mor}_p(\text{appr}(x, k)).$$

3 Proof

Lemma 3 (square roots of 1; `sq_eq_one_zmod`). *If p is odd and $x^2 = 1$ in $\mathbb{Z}/p^k$, then $x = \pm 1$.*

Proof. Lift x to $y \in \mathbb{Z}$. Then $p^k \mid (y-1)(y+1)$. The prime p cannot divide both factors, since it would divide their difference 2. So p^k is coprime to one factor and divides the other. $\square$

Lemma 4 (generalized Wilson theorem; `prod_units_eq_neg_one`, `prod_Ico_fac`). *For odd p and $k \geq 1$, the product of all units of $\mathbb{Z}/p^k$ is -1 . Hence, for every n , $\prod_{n \leq j < n+p^k, p \nmid j} j \equiv -1 \pmod{p^k}$.*

Proof. Pair each unit u with u^{-1} . By the previous lemma, only ± 1 are their own inverses. So the product over the units other than -1 is 1, and the full product is -1 . The units of $\mathbb{Z}/p^k$ are exactly the residues of the $j \in [0, p^k)$ with $p \nmid j$. This gives the window $n = 0$. Moving the window from $[n, n+p^k)$ to $[n+1, n+1+p^k)$ removes the factor for $j = n$ and adds the factor for $j = n+p^k$. These factors are congruent, and each is a unit mod p^k or equal to 1. So every window has the same product. $\square$

Lemma 5 (Morita congruence; `mor_add_period`, `mor_congr`). *For odd p : if $n \equiv m \pmod{p^k}$ then $\text{mor}_p(n) \equiv \text{mor}_p(m) \pmod{p^k}$.*

Proof. $\text{mor}_p(n+p^k) = \text{mor}_p(n) \cdot (-1)^{p^k} \cdot \prod_{n \leq j < n+p^k, p \nmid j} j \equiv \text{mor}_p(n) \cdot (-1) \cdot (-1)$, because p^k is odd. Then induct on $(m-n)/p^k$. $\square$

Theorem 6 (construction; `cauchySeq_approx`, `toZModPow_gammaP`, `gammaP_natCast`, `continuous_gammaP`, `gammaP_unique`). *Let p be an odd prime. The limit defining $\Gamma_p(x)$ exists. For all x and k , $\Gamma_p(x) \equiv \text{mor}_p(\text{appr}(x, k)) \pmod{p^k}$, and hence $\Gamma_p(x) \equiv \text{mor}_p(n) \pmod{p^k}$ whenever $x \equiv n \pmod{p^k}$ with $n \in \mathbb{N}$. Moreover $\Gamma_p(n) = \text{mor}_p(n)$ for all $n \in \mathbb{N}$, and Γ_p is continuous. Every continuous $G : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ with $G(n) = \text{mor}_p(n)$ for all positive integers n equals Γ_p . So Γ_p is Morita's function.*

Proof. Since $\text{appr}(x, k+1) \equiv \text{appr}(x, k) \pmod{p^k}$, the Morita congruence gives $|\text{mor}_p(\text{appr}(x, k+1)) - \text{mor}_p(\text{appr}(x, k))|_p \leq p^{-k}$. So the sequence is Cauchy, and it converges because $\mathbb{Z}_p$ is complete. The same bound for all $m \geq k$ passes to the limit, which gives the congruence mod p^k . For $x = n \in \mathbb{N}$, these congruences hold for every k , so $\Gamma_p(n) = \text{mor}_p(n)$. If $|x - y|_p \leq p^{-k}$, then $\Gamma_p(x) \equiv \Gamma_p(y) \pmod{p^k}$, which gives continuity. Finally, the positive integers are dense in $\mathbb{Z}_p$, and two continuous functions that agree on a dense set are equal. $\square$

Theorem 7 (reading (A); `reflection_products`, `reflection_product_not_constant`, `not_function_of_p_mod_four`). *For every odd prime p : $\Gamma_p(1)\Gamma_p(1-1) = -1$, while $\Gamma_p(2)\Gamma_p(1-2) \equiv 1 \pmod{p}$. Hence $a \mapsto \Gamma_p(a)\Gamma_p(1-a)$ is not constant on $\mathbb{Z}_p$, and no function f of $p \bmod 4$ satisfies (A).*

Proof. We have $\Gamma_p(0) = \text{mor}_p(0) = 1$, $\Gamma_p(1) = \text{mor}_p(1) = -1$ and $\Gamma_p(2) = \text{mor}_p(2) = 1$. Since $-1 \equiv p-1 \pmod{p}$, the construction theorem gives $\Gamma_p(-1) \equiv \text{mor}_p(p-1) \pmod{p}$. Also $\text{mor}_p(p) = \text{mor}_p(p-1) \cdot (-1) \cdot (p-1) \equiv \text{mor}_p(p-1) \pmod{p}$, and $\text{mor}_p(p) \equiv \text{mor}_p(0) = 1 \pmod{p}$ by the Morita congruence. So $\Gamma_p(2)\Gamma_p(-1) \equiv 1$, while $\Gamma_p(1)\Gamma_p(0) = -1$. Since $p > 2$, $1 \not\equiv -1 \pmod{p}$. If f existed, then for $p = 3$ both products would equal $f(3)$, a contradiction. $\square$

Theorem 8 (reading (B); `reflection_four`, `not_function_of_p_mod_four_at_four`). *$\Gamma_3(4)\Gamma_3(-3) \equiv -1 \pmod{3}$ and $\Gamma_7(4)\Gamma_7(-3) \equiv 1 \pmod{7}$. Hence there is no $s : \mathbb{Z}/4 \rightarrow \{\pm 1\}$ with $\Gamma_p(4)\Gamma_p(1-4) = s(p \bmod 4)$ for all odd primes p .*

Proof. $\Gamma_3(4) = \text{mor}_3(4) = 1 \cdot 2 = 2$, and $-3 \equiv 0 \pmod{3}$, so $\Gamma_3(-3) \equiv \text{mor}_3(0) = 1$. The product is $\equiv 2 \equiv -1 \pmod{3}$. Next, $\Gamma_7(4) = \text{mor}_7(4) = 6$, and $-3 \equiv 4 \pmod{7}$, so $\Gamma_7(-3) \equiv 6$. The product is $\equiv 36 \equiv 1 \pmod{7}$. Since $3 \equiv 7 \pmod{4}$, $s(3)$ would be $\equiv -1 \pmod{3}$ and $\equiv 1 \pmod{7}$. Neither $+1$ nor -1 satisfies both. $\square$

4 Lean formalization

The file `lean/Conjecture716/Basic.lean` (Lean 4, Mathlib v4.33.1, `import Mathlib` only) defines `C716.mor`, `C716.approx` and `C716.gammaP`, and proves all the results above under the names given in brackets. The two main statements are:

```
theorem not_function_of_p_mod_four :
  not exists f : ZMod 4 -> Int, forall (p : Nat) [Fact p.Prime], p != 2 ->
    forall a : Z_[p], gammaP p a * gammaP p (1 - a) = ((f (p : ZMod 4) : Int) : Z_[p])
theorem not_function_of_p_mod_four_at_four :
  not exists s : ZMod 4 -> Int^x, forall (p : Nat) [Fact p.Prime], p != 2 ->
    gammaP p 4 * gammaP p (1 - 4) = (((s (p : ZMod 4) : Int^x) : Int) : Z_[p])
```

(ASCII transcription of the Lean source: `not`, `exists`, `forall`, `->` and `!=` stand for the Lean symbols for negation, the quantifiers, the arrow and $\neq$; `Int`, `Nat`, `Z_[p]` and `Int^x` stand for $\mathbb{Z}$, $\mathbb{N}$, the p -adic integers and the units $\mathbb{Z}^\times = \{\pm 1\}$.) `reflection_product_not_constant` proves that for every odd prime p , there is no $c \in \mathbb{Z}_p$ with $\Gamma_p(a)\Gamma_p(1-a) = c$ for all a . `gammaP_unique` proves that `gammaP p` is the unique continuous function interpolating mor_p on the positive integers.

Conventions. The primes are odd ($p \neq 2$), and $\mathbb{N}$ contains 0. Congruences mod p^k are stated in Lean through the ring map `PadicInt.toZModPow k`. The “sign” in reading (B) is a unit of $\mathbb{Z}$, i.e. ± 1 . In reading (A), f may take any integer value.

Axiom audit. `#print axioms` for both main theorems, and for `gammaP_unique`, `continuous_gammaP` and `gammaP_natCast`, reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`. The finite checks in $\mathbb{Z}/3$ and $\mathbb{Z}/7$ use the kernel `decide`.